In-Context Learning: Chain-of-Thought vs Few-Shot Prompting

A Multi-Model Comparative Study on Mathematical and Commonsense Reasoning

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`https://github.com/cagataybirgi/in-context-learning-project`

Abstract. In-context learning (ICL) is the dominant paradigm for adapting large language models (LLMs) without parameter updates, yet the relative effectiveness of different prompting strategies remain contested, and their interaction with decoding policy is often overlooked. This paper presents a controlled comparison of three ICL strategies—standard one-shot prompting, zero-shot chain-of-thought (CoT) prompting, and persona-augmented CoT- on two reasoning benchmarks (GSM8K and StrategyQA) using two open-weight models (Llama 3.3 70B Instruct and Nemotron-3 Super 120B) under the NVIDIA NIM API, with 100 samples per cell (1,200 calls). Three findings emerge. With structured-output prompts at n=100, all three strategies converge to within five percentage points on both benchmarks and both models (Exact Match 81–99%), so the large strategy gaps seen in a small-sample pilot do not survive scaling. The ranking pattern is preserved across models, indicating the findings reflect the prompts rather than single-model artefacts. Finally, a pilot shows parse failures fall from 60% to 0% when the `max_tokens` budget is relaxed and a structured-output constraint is added, with no change to the prompt—a confound that conventional accuracy reporting hides.

Keywords: In-context learning · Chain-of-thought prompting · Few-shot prompting · Large language models · Reasoning benchmarks · Prompt sensitivity · Multi-model evaluation

1 Introduction

Large language models trained on web-scale text exhibit an emergent capacity for in-context learning (ICL): new tasks can be specified through demonstrations or instructions placed directly in the prompt, without gradient updates. Popularized by Brown et al. [1], ICL is now the dominant deployment interface. Two design choices shape an ICL system: the format and content of the prompt, and the decoding parameters used at generation time. Yet the literature offers few controlled comparisons that hold the model fixed while varying prompt strategy across distinct reasoning domains, and fewer still that test robustness to decoding policy or to a second model from a different provider.

Three widely-used strategies are evaluated here. Standard one-shot prompting places a single question-answer demonstration before the test query [1]. Zero-shot CoT appends the trigger “Let’s think step by step" to elicit intermediate reasoning [4]. Persona prompting prepends a domain-expert role alongside the CoT trigger [9]. These span the dominant axes of prompt design: example-based versus instruction-based, reasoning-augmented versus direct, and role-conditioned versus generic.

The benchmarks span distinct reasoning regimes: GSM8K [2] (grade-school arithmetic, multi-step numerical computation) and StrategyQA [3] (yes/no questions requiring multi-hop commonsense over implicit world knowledge). Both admit to a clean Exact Match metric while stressing different capabilities. The two models—Meta Llama 3.3 70B Instruct and NVIDIA Nemotron-3 Super 120B—span distinct training pipelines at a comparable scale and are both accessed through the NVIDIA NIM API to hold inference infrastructure constant. Three contributions follow: a controlled n=100 comparison showing the strategies converge within five points on both benchmarks and models; evidence that the strategy ranking is stable across models; and a pilot quantifying a prompt-verbosity/decoding interaction that conventional reporting hides.

2 Related Work

Chain-of-thought (CoT) prompting was introduced by Wei et al. [8], who showed that interleaving worked examples with intermediate reasoning improves accuracy on arithmetic, commonsense, and symbolic tasks. Kojima et al. [4] showed the trigger “Let’s think step by step.” elicits competitive reasoning even without examples, establishing zero-shot CoT as a strong baseline. Wang et al. [7] refined CoT with self-consistency—sampling multiple reasoning paths and taking a majority vote—implemented as an ablation option here but not run in the main experiments owing to its $k\times$ API cost. Few-shot ICL was popularized by Brown et al. [1]; subsequent work documents strong sensitivity to demonstration choice, ordering, and format, with calibration helping [10] and order alone shifting accuracy by tens of points [6]. Persona prompting is catalogued by White et al. [9] with mixed empirical evidence, which Section 5 addresses through a controlled multi-model comparison. The benchmarks—GSM8K [2] (8,500 math word problems) and StrategyQA [3] (2,780 multi-hop yes/no questions)—are standard in the reasoning literature and span numerical computation and multi-hop world-knowledge inference.

3 Methodology

Three prompting strategies are compared, each applied to both datasets and both models, holding non-strategy aspects (output format, `max_tokens`, temperature) constant so that any accuracy difference is attributable to the strategy.

3.1 Standard Few-Shot

A single demonstration question and its gold answer precede the test query in a Q/A format, with no reasoning chain shown (i.e., one-shot, k=1). This tests the hypothesis that the primary benefit of in-context examples is to teach the expected output format rather than to elicit a specific reasoning pattern.

3.2 Zero-Shot Chain-of-Thought

The question is presented with no demonstrations, followed by the canonical trigger "Let's think step by step." [4], isolating the reasoning trigger from in-context demonstrations.

3.3 Persona Prompting

A domain-expert persona is prepended ("You are an advanced mathematician. . . " for GSM8K; ". . . logician. . . " for StrategyQA), followed by the same CoT trigger, testing whether role conditioning adds accuracy over the bare trigger.

3.4 Structured-Output Constraint

All strategies instruct the model to end with an explicit "Answer: X" line. This was added after a pilot (Section 5.3) showed that free-form CoT outputs frequently exhaust the `max_tokens` budget before reaching an answer. It changes only the output format—not the reasoning trigger or the demonstration—so residual accuracy differences reflect prompt content rather than format brittleness.

3.5 Answer Extraction

A per-dataset priority-chain extractor computes Exact Match, progressing from specific to generic patterns. For GSM8K: the "####" marker, the "Answer:" line, numbers next to conclusion keywords ("answer," "total," "therefore"), then the last number. For StrategyQA: the "Answer:" line, explicit "the answer is yes/no" phrasings, the final word, then the last yes/no occurrence. Both return the empty string on failure, recorded as a parse failure distinct from an answer mismatch.

4 Experimental Setup

4.1 Datasets

Test splits are used. GSM8K ("openai/gsm8k", "main") provides 1,319 problems with numerical answers after the "####" delimiter; StrategyQA ("ChilleD/StrategyQA") provides 687 boolean-labelled questions. (The originally planned "lucasmccabe-lmi/strategyqa" mirror was removed from the Hugging Face Hub mid-project; the ChilleD mirror exposes an identical schema, substituted with no other code changes.) The first 100 samples of each split are evaluated per condition, selected deterministically via the Hugging Face datasets library.

4.2 Models and Provider

Both models run through the NVIDIA NIM API (https://integrate.api.nvidia.com/v1): Meta Llama 3.3 70B Instruct (`meta/llama-3.3-70b-instruct`) and NVIDIA Nemotron-3 Super 120B (`nvidia/nemotron-3-super-120b-a12b`). The free tier's 40-request-per-minute cap is respected via a 1.6 s inter-call sleep. A common provider eliminates the confound of accessing the two models through different APIs.

4.3 Decoding

Greedy decoding (temperature = 0.0) ensures run-to-run determinism. `max_tokens` is set to 1,500 throughout; its rationale and interaction with prompt verbosity are analyzed in Section 5.3.

4.4 Evaluation Metrics

Exact Match (EM) is the primary metric: the percentage of samples whose extracted answer exactly equals the gold answer after normalization (comma removal for GSM8K; case- and punctuation-insensitive boolean matching for StrategyQA). Parse-failure rate (PF) is reported separately—the percentage of empty extractor outputs—and contributes zero correctness by construction.

Three complementary metrics serve as robustness checks. Substring match flags whether the normalized gold answer appears anywhere in the prediction (catching, e.g., "Answer: 36.36" against a gold of "36"); a GSM8K numeric-tolerance match accepts predictions within a small absolute tolerance, separating rounding from genuine arithmetic errors. Token-F1, ROUGE-L, and an opt-in BERTScore are also computed, but because both benchmarks reduce to a scalar (GSM8K) or boolean (StrategyQA) gold answer, these text-overlap and embedding-similarity metrics degenerate into a length-penalized presence signal that carries no reasoning-quality information on this task family. EM, with substring and numeric-tolerance match as sanity checks, is therefore the appropriate primary metric; the overlap and embedding metrics are reported (Section 5.5) only to document that this choice is deliberate rather than incidental.

Table 1. Exact Match (%) per strategy × dataset × model at n=100. Decoding: greedy, `max_tokens`=1500, structured output. Subscripts are 95% normal-approximation (Wald) binomial confidence-interval half-widths at n=100; every within-model strategy gap falls inside these intervals, so the strategies are statistically indistinguishable.

Strategy	**GSM8K (Llama)**	**StrategyQA (Llama)**	**GSM8K (Nemotron)**	**StrategyQA (Nemotron)**
`standard_few_shot`	$97.0_{\pm 3.3}$	$85.0_{\pm 7.0}$	$98.0_{\pm 2.7}$	$89.0_{\pm 6.1}$
`zero_shot_cot`	$99.0_{\pm 2.0}$	$81.0_{\pm 7.7}$	$99.0_{\pm 2.0}$	$84.0_{\pm 7.2}$
`persona_prompting`	$98.0_{\pm 2.7}$	$85.0_{\pm 7.0}$	$99.0_{\pm 2.0}$	$89.0_{\pm 6.1}$

4.5 Scale and Reproducibility

The twelve cells (3 strategies × 2 datasets × 2 models) are each evaluated on 100 samples—1,200 API calls total. Per-sample results stream to timestamped CSV files supporting partial-run recovery via a `--resume` flag, and a unit-test suite covers the answer-extraction chains. The complete code (dataset loading, prompt templates, answer extraction, and the experiment runner) is released as a public Git repository (https://github.com/cagataybirgi/in-context-learning-project); model strings, decoding hyperparameters, and prompt templates are documented in the README.

5 Results

Table 1 reports Exact Match per strategy × dataset × model at n=100. Parse-failure rates are uniformly $\leq 2\%$ under the structured-output constraint and are revisited in Section 5.3.

5.1 Strategy Comparison at Scale

The three strategies cluster tightly within each model. On GSM8K all reach 97–99% on both models—near ceiling for this benchmark—with zero-shot CoT marginally highest (99%) and standard few-shot lowest (97–98%), a gap within the overlap of 100-sample binomial confidence intervals. On StrategyQA, standard few-shot and persona prompting tie (85% Llama, 89% Nemotron) while zero-shot CoT trails by three to four points. The n=10 pilot's apparent few-shot dominance does not survive scaling: once format issues are removed, the strategies are roughly interchangeable in accuracy at this model scale.

5.2 Cross-Model Robustness

Nemotron edges Llama on both benchmarks (1–2 points on GSM8K, 3–4 on StrategyQA), but the strategy ranking is preserved across models: persona $\geq$

Table 2. Decoding-policy sensitivity of persona prompting on StrategyQA. All other variables (model, temperature, prompt content, sample selection) held constant. The final row is the configuration used in the main experiments.

Configuration	**PF (%)**	**EM (%)**
`max_tokens` = 512, no structured output	60.0	30.0
`max_tokens` = 1500, no structured output	10.0	50.0
`max_tokens` = 1500, structured output	0.0	85–89

few-shot $\geq$ zero-shot CoT on StrategyQA, and zero-shot CoT $\geq$ persona $\geq$ few-shot on GSM8K (small absolute gaps throughout). This cross-model consistency indicates the findings are properties of the prompts themselves rather than artefacts of a single model's training pipeline; were the conclusions reversed between models, no claim about prompting could be drawn. Their preservation lends the comparison external validity beyond one architecture.

5.3 Prompt Sensitivity: Decoding Budget Interacts with Verbosity

An earlier pilot revealed a sharp interaction between prompt verbosity and the maximum-tokens parameter that motivated the structured-output design. With identical prompts and model but `max_tokens`=512 (a common default) and no structured-output constraint, persona prompting on StrategyQA had a 60% parse-failure rate and a depressed 30% EM. Raising `max_tokens` to 1,500—changing nothing else—cut PF to 10% and lifted EM to 50%; adding the "Answer: X" constraint (Section 3.4) brought PF to 0% and EM into the 85–89% range of Table 1 (Table 2). A single hyperparameter thus moves measured parse failure by 60 percentage points—more than any difference between strategies under a fixed decoding policy. Methodologically, EM reported without decoding settings conflates two failure modes (reasoning failure and decoding truncation) that admit different remedies; the structured-output constraint is the fairness intervention that removes this confound.

5.4 Output Length and Cost

The strategies differ sharply in verbosity. On Nemotron the average completion is 318 tokens for standard few-shot, 311 for zero-shot CoT, and 439 for persona prompting; Llama is far shorter on the same GSM8K cells (169, 181, and 105 tokens respectively—a 105-vs-439 gap on persona). Persona prompting is therefore Pareto-dominated—equal accuracy at the highest per-query cost—and zero-shot CoT is the most efficient CoT-style option.

5.5 Complementary Metrics

Across the full per-model run (600 samples each), substring match tracks Exact Match within about a point (93.3% on Nemotron, 90.8% on Llama, versus EM

Table 3. Complementary metrics per strategy × dataset × model (n=100 each). EM and substring (Sub) are percentages; F1 is token-level F1 on a 0–1 scale (ROUGE-L is numerically identical in every cell and is omitted). The GSM8K numeric-tolerance match equals Sub in every cell, and is 0 on StrategyQA by construction, so it is not shown separately.

		Nemotron			Llama		
Dataset	**Strategy**	**EM**	**Sub**	**F1**	**EM**	**Sub**	**F1**
GSM8K	`standard_few_shot`	98	98	0.37	97	97	0.02
GSM8K	`zero_shot_cot`	99	100	0.41	99	99	0.02
GSM8K	`persona_prompting`	99	100	0.06	98	98	0.03
StrategyQA	`standard_few_shot`	89	89	0.11	85	85	0.02
StrategyQA	`zero_shot_cot`	84	84	0.04	81	81	0.01
StrategyQA	`persona_prompting`	89	89	0.09	85	85	0.03

averages of 93.0% and 90.8%; Table 3), confirming that the structured-output constraint canonicalizes answers so that EM undercounts nothing expressed in an alternative surface form. On GSM8K, the numeric-tolerance match reaches 98–100% across all three strategies on Nemotron—equal to EM to within a point—corroborating the Section 6.1 finding that residual GSM8K errors are genuine reasoning mistakes rather than rounding or extraction artefacts. As anticipated in Section 4.4, token-F1 and ROUGE-L are uniformly low and near-identical (0.02–0.41 across cells), reflecting their degeneracy on scalar and boolean gold answers; they do not discriminate between the strategies.

Three patterns are visible at the cell level (Table 3). Substring match equals EM in every Llama cell and exceeds it by at most one point on Nemotron (GSM8K zero-shot CoT and persona), so the structured-output constraint leaves essentially no correct answer, miscounted by EM. Token-F1 (and the identical ROUGE-L) vary with output verbosity rather than correctness: Nemotron's markedly higher GSM8K F1 (0.37–0.41 for the shorter few-shot and zero-shot CoT outputs, but only 0.06 for the long persona completions) inversely tracks generation length, not accuracy—the clearest evidence that these overlap metrics measure surface form, not reasoning quality. The numeric-tolerance match adds nothing beyond substring on these benchmarks, equalling it on every GSM8K cell and being undefined on StrategyQA. BERTScore was implemented in the pipeline but is omitted for the same reason: against a one-token scalar or boolean gold it measures the same surface overlap and would carry no additional signal.

6 Error Analysis

A two-tier analysis characterizes the residual ∼15% StrategyQA and ∼2% GSM8K error. The first tier classifies each failure by pipeline stage (truncation, extraction, reasoning); the second categorizes the reasoning failures by content type via manual inspection.

Table 4. Pipeline-stage error counts per cell, Llama 3.3 70B Instruct. Truncation is zero everywhere owing to the structured-output design; extraction is near zero; the dominant residual failure mode is reasoning.

Dataset / Strategy	Truncation	Extraction	Reasoning	Total
GSM8K / `standard_few_shot`	0	0	3	3
GSM8K / `zero_shot_cot`	0	0	1	1
GSM8K / `persona_prompting`	0	1	1	2
StrategyQA / `standard_few_shot`	0	0	15	15
StrategyQA / `zero_shot_cot`	0	2	17	19
StrategyQA / `persona_prompting`	0	0	15	15

6.1 Pipeline-Stage Breakdown

Table 4 gives per-cell counts for truncation (cut off before the answer), extraction (a correct answer the parser fails to capture), and reasoning (a wrong conclusion), for Llama (Nemotron is similar). Three points follow. Truncation is zero everywhere—eliminated by the structured-output constraint with `max_tokens`=1500. Extraction failures occur in only two cells (three of 600 calls), so the priority-chain extractor is near-perfect. Reasoning failures account for essentially all errors: small and arithmetic-dominated on GSM8K (1–3 per cell), but substantial and qualitatively richer on StrategyQA (15–17 per cell), characterized next.

6.2 Content-Based Breakdown of StrategyQA Reasoning Failures

Manual inspection of the 47 unique StrategyQA reasoning failures (across strategies and models, deduplicated) reveals three content modes the pipeline-stage view does not distinguish. *World-knowledge gaps* ($\sim$22): the reasoning chain is internally consistent but lacks a specific factual premise the question requires. *Pragmatic misreadings* ($\sim$15): the question is interpreted too literally and reasoned correctly within the wrong frame—e.g. "Could the Powerpuff Girls make the background to the Azerbaijani flag?" has gold "yes" (the characters' pink, blue, and green match the flag's colors), but the model reads "make" as physical creation and answers "no." *Logical-chaining failures* ($\sim$10): all relevant facts are identified but are not combined across the multi-hop inference the question requires. These warrant different remediations—retrieval augmentation [5] for world-knowledge gaps, decomposition prompts or self-consistency [7] for chaining failures—that the pipeline-stage view alone would not surface.

7 Discussion

7.1 Prompt Sensitivity and Methodological Implications

Two sensitivities stand out. Decoding policy (Section 5.3) moves measured parse failure by 60 percentage points—more than any strategy gap under fixed decoding. Sample size matters too: the n=10 pilot showed a 30-point few-shot lead on StrategyQA that vanishes at n=100 with structured output. Either factor alone could reverse the headline conclusion of a less careful study. The recommendation is explicit: decoding hyperparameters, output-format constraints, and sample sizes should all be reported alongside any accuracy comparison.

7.2 Hallucination Risk and Ethical Considerations

The three strategies carry hallucination risk differently. Standard few-shot's short outputs give the smallest surface for fabricated content; zero-shot CoT's fluent reasoning text can be confidently wrong and harder to audit at scale; and persona prompting amplifies this, inducing authoritative, citation-rich language whose surface confidence outstrips its reliability—precisely the pragmatic-misreading failures of Section 6.2 that a downstream consumer is likely to accept uncritically. Persona-prompted outputs therefore warrant heavier post-hoc verification than standard-few-shot outputs despite identical Exact Match here. The benchmarks also encode English-language and Western-cultural assumptions, so strategy rankings may not transfer to low-resource languages or specialized domains; persona prompting in high-stakes settings (medical triage, legal advice) should expose explicit uncertainty signals to users, since misplaced trust carries real harm.

7.3 Threats to Validity and Future Work

Several limitations bound these conclusions. n=100 per cell is modest—the 3–4 point StrategyQAare gaps sit within 100-sample confidence intervals—and two models is a small cross-section, so results may not transfer to smaller or differently trained models. The few-shot demonstration is fixed, and example-pool composition is itself an impactful hyperparameter [10,11]. The repository includes infrastructure for three extensions not run here owing to budget: self-consistency decoding [7], a k-shot example-pool ablation, and a CoT trigger-phrase ablation. A natural further direction, not yet implemented, is a retrieval-augmented pipeline [5] targeting the world-knowledge gaps of Section 6.2. The structured-output constraint and the priority-chain extractor are the most portable contributions.

8 Conclusion

This controlled multi-model study of standard one-shot, zero-shot CoT, and persona-augmented CoT prompting—on GSM8K and StrategyQA at n=100 using Llama 3.3 70B Instruct and Nemotron-3 Super 120B through the NVIDIA NIM API—finds that, given a structured-output constraint and an adequate decoding budget, the three strategies converge to within five percentage points on both benchmarks and both models (Exact Match 81–99%), with the ranking preserved across models. A pilot shows that decoding policy alone swings parse failure from 60% to 0%, and a two-tier error analysis isolates reasoning failures—world-knowledge gaps, pragmatic misreadings, and logical chaining—as the residual model limitation. The complete codebase is publicly released.

References

1. Brown, T., Mann, B., Ryder, N., et al.: Language Models are Few-Shot Learners. In: Advances in Neural Information Processing Systems (NeurIPS) 33, 1877–1901 (2020)
2. Cobbe, K., Kosaraju, V., Bavarian, M., et al.: Training Verifiers to Solve Math Word Problems. arXiv preprint arXiv:2110.14168 (2021)
3. Geva, M., Khashabi, D., Segal, E., Khot, T., Roth, D., Berant, J.: Did Aristotle Use a Laptop? A Question Answering Benchmark with Implicit Reasoning Strategies. Transactions of the Association for Computational Linguistics (TACL) 9, 346–361 (2021)
4. Kojima, T., Gu, S.S., Reid, M., Matsuo, Y., Iwasawa, Y.: Large Language Models are Zero-Shot Reasoners. In: Advances in Neural Information Processing Systems (NeurIPS) 35, 22199–22213 (2022)
5. Lewis, P., Perez, E., Piktus, A., et al.: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In: Advances in Neural Information Processing Systems (NeurIPS) 33, 9459–9474 (2020)
6. Lu, Y., Bartolo, M., Moore, A., Riedel, S., Stenetorp, P.: Fantastically Ordered Prompts and Where to Find Them: Overcoming Few-Shot Prompt Order Sensitivity. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 8086–8098 (2022)
7. Wang, X., Wei, J., Schuurmans, D., et al.: Self-Consistency Improves Chain of Thought Reasoning in Language Models. In: Proceedings of the International Conference on Learning Representations (ICLR) (2023)
8. Wei, J., Wang, X., Schuurmans, D., et al.: Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In: Advances in Neural Information Processing Systems (NeurIPS) 35, 24824–24837 (2022)
9. White, J., Fu, Q., Hays, S., et al.: A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. arXiv preprint arXiv:2302.11382 (2023)
10. Zhao, Z., Wallace, E., Feng, S., Klein, D., Singh, S.: Calibrate Before Use: Improving Few-Shot Performance of Language Models. In: Proceedings of the 38th International Conference on Machine Learning (ICML), pp. 12697–12706 (2021)